

differentiation, and translation properties of the Laplace transformation.

- Proper rational functions with simple poles can be expanded by partial fraction to obtain inverse Laplace transforms. Simple real poles lead to exponential waveforms and simple complex poles to damped sinusoids. Partial-fraction expansions of improper rational functions and functions with multiple poles require special treatment. MATLAB software can be used to find Laplace transforms from $f(t)$ and waveforms from $F(s)$.

- Using Laplace transforms to find the response of a circuit involves transforming the circuit differential equation into the s domain, algebraically solving for the response transform, and performing the inverse transformation to obtain the response waveform.
- The initial and final value properties determine the initial and final values of a waveform $f(t)$ from the value of $sF(s)$ as $s \rightarrow \infty$ and $s \rightarrow 0$, respectively. The initial value property applies if $F(s)$ is a proper rational function. The final value property applies if the poles of $sF(s)$ are in the left half plane.

PROBLEMS

OBJECTIVE 9-1 LAPLACE TRANSFORM (SECTS. 9-1, 9-2, 9-3)

Find the Laplace transform of a given signal waveform using transform properties and pairs, using the integral definition of the Laplace transformation, or using software applications. Locate the poles and zeros of the transform and construct a pole-zero diagram.

See Examples 9-1, 9-2, 9-3, 9-4, 9-5, 9-6, 9-7, 9-8, 9-9, 9-10 and Exercises 9-1, 9-2, 9-3, 9-4, 9-5, 9-6, 9-7, 9-8, 9-9, 9-10, 9-11, 9-12

- 9-1 Find the Laplace transform of $f(t) = 10[e^{-50t} - 2e^{-100t}]u(t)$. Locate the poles and zeros of $F(s)$.

- 9-2 Find the Laplace transform of $f(t) = A[(3 - \alpha t)e^{-\alpha t}]u(t)$. Locate the poles and zeros of $F(s)$.

- 9-3 Find the Laplace transform of $f(t) = 25[1 - 3 \cos(500t)]u(t)$. Locate the poles and zeros of $F(s)$.

- 9-4 Find the Laplace transform of $f(t) = A[\cos(\beta t) - \sin(\beta t)]u(t)$. Locate the poles and zeros of $F(s)$.

- 9-5 Find the Laplace transform of $f(t) = 28(t - 100e^{-10t} \cos(t))u(10t)$. Locate the poles and zeros of $F(s)$.

- 9-6 Find the Laplace transform of $f(t) = 30[2 - 5t - 2e^{-5t}]u(t)$. Locate the poles and zeros of $F(s)$.

- 9-7 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams:

- (a) $f_1(t) = [15e^{-5t} - 20e^{-10t}]u(t)$
(b) $f_2(t) = [5 \cos 10t + 10 \cos 20t]u(t)$

- 9-8 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams.

- (a) $f_1(t) = 48(t) + [20e^{-20t} + 40e^{-40t}]u(t)$
(b) $f_2(t) = [15 + 15 \cos 500t]u(t)$

- 9-9 Use MATLAB to find the Laplace transforms of the following waveforms. Locate the poles and zeros of $F(s)$.

- (a) $f_1(t) = 8(t) + (625te^{-25t})u(t)$
(b) $f_2(t) = [10 + e^{-10t} + 5 \cos 10t + 5 \sin 10t]u(t)$

- 9-10 Find the Laplace transforms of the following waveforms:

- (a) $f_1(t) = 38(t - 2)$
(b) $f_2(t) = 10e^{-50(t-1)}u(t - 1)$
(c) $f_3(t) = 10e^{-50(t-2)}u(t - 2)$

- 9-11 Use MATLAB to find the Laplace transform of the following waveform.

$$f(t) = [10 + 2e^{-10t}]u(t) + [5 \cos 100(t - 0.05)]u(t - 0.05)$$

- 9-12 Find the Laplace transform of the following waveform:

- (a) $f_1(t) = \frac{d}{dt}(100e^{-5000t} \cos 20,000t)u(t)$
(b) $f_2(t) = \int_0^t 50e^{-50x} dx + 100u(t) + 5 \frac{de^{-50t}}{dt}u(t)$

- 9-13 (a) Write an expression for the waveform $f(t)$ in Fig. P9-13 using step and ramp functions.

- (b) Use the time-domain translation property to find the Laplace transform of the waveform $f(t)$ found in part (a).

- (c) Verify the Laplace transform found in part (b) by applying the definition of the Laplace transform to the waveform $f(t)$ found in part (a).

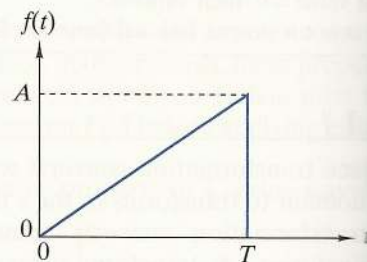


FIGURE P9-13

- (a) Write an expression for the waveform $f(t)$ in Fig. P9-14 using step functions.
 (b) Use the time-domain translation property to find the Laplace transform of the waveform $f(t)$ found in part (a).
 (c) Verify the Laplace transform found in part (b) by applying the definition of the Laplace transformation in Eq. (9-2) to the waveform $f(t)$ found in part (a).

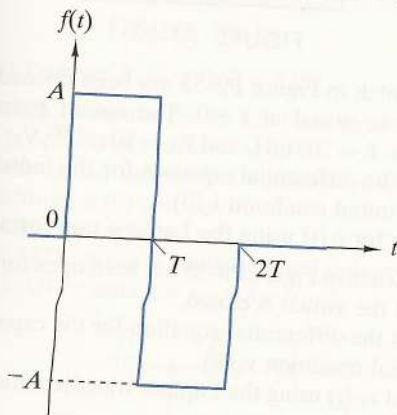


FIGURE P9-14

- 9-15 (a) Find the Laplace transform of the following function. Locate the poles and zeros of $F(s)$.

$$f(t) = [500 + 75e^{-200t}t \sin 200t]u(t)$$

 (b) Validate your result using MATLAB.

OBJECTIVE 9-2 INVERSE TRANSFORMS SECTS. 9-4, 9-5

Find the inverse transform of a given Laplace transform using partial fraction expansion, using basic transform properties and pairs, or using software applications.
 Examples 9-11, 9-12, 9-13, 9-14 and Exercises 9-13, 9-14, 9-15, 9-16, 9-17, 9-18, 9-19, 9-20, 9-21

- 9-16 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{s+100}{s(s+50)}$
 (b) $F_2(s) = \frac{s+10}{s(s+50)(s+100)}$

- 9-17 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{50(s+10)}{(s+5)(s+20)}$
 (b) $F_2(s) = \frac{2s^2}{(s+200)(s+500)}$

- 9-18 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = 400 \frac{1}{(s+10)^2 + 20^2}$

(b) $F_2(s) = \frac{2(s+10)}{(s+10)^2 + 20^2}$

- 9-19 Find the inverse Laplace transforms of the following functions and sketch their waveforms for $\beta > 0$:

(a) $F_1(s) = \frac{\beta(s+\beta)}{s(s^2 + \beta^2)}$

(b) $F_2(s) = \frac{s(s+\beta)}{s^2 + \beta^2}$

- 9-20 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{\alpha^2}{s^2(s+\alpha)}$

(b) $F_2(s) = \frac{\alpha^2}{s(s+\alpha)^2}$

- 9-21 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{600}{(s+10)(s+20)(s+40)}$

(b) $F_2(s) = \frac{6(s+10)}{(s+5)(s+20)}$

- 9-22 Use MATLAB to find the inverse transforms of the following functions:

(a) $F_1(s) = \frac{16s}{(s+2)(s^2 + 12s + 13)}$

(b) $F_2(s) = \frac{4(s^2 + 4)}{s(s^2 + 16)}$

- 9-23 Find the inverse transforms of the following functions:

(a) $F_1(s) = \frac{(s+4000)(s+8000)}{s(s+2000)(s+6000)}$

(b) $F_2(s) = \frac{3s^4 + 10s^2 + 4}{s(s^2 + 1)(s^2 + 4)}$

- 9-24 Find the inverse transforms of the following functions:

(a) $F_1(s) = \frac{300(s+20)}{s(s^2 + 40s + 300)}$

(b) $F_2(s) = \frac{200s}{(s+40)(s^2 + 4s + 8)}$

- 9-25 Find the inverse transforms of the following functions:

(a) $F_1(s) = \frac{4(s^2 + 16)}{s(s^2 + 8s + 32)}$

(b) $F_2(s) = \frac{2(s^2 + 30s + 800)}{s(s^2 + 50s + 400)}$

- 9-26 Find the inverse transforms of the following functions and validate your answers using MATLAB:

(a) $F_1(s) = \frac{(s+50)^2}{(s+20)^2(s+100)}$

(b) $F_2(s) = \frac{(s+20)^2}{(s+50)^2(s+100)}$

9-27 A certain transform has a simple pole at $s = -20$, a simple zero at $s = -\gamma$, and a scale factor of $K = 1$. Select values for γ so the inverse transform is

- (a) $f(t) = \delta(t) - 5e^{-20t}$
 (b) $f(t) = \delta(t)$
 (c) $f(t) = \delta(t) + 5e^{-20t}$

9-28 Find the inverse transforms of the following functions:

- (a) $F_1(s) = \frac{e^{-3s}(s+20)}{(s+10)(s+30)}$
 (b) $F_2(s) = \frac{se^{-3s} + 20}{(s+10)(s+30)}$

9-29 Use MATLAB to find the inverse transform and plot the poles and zeros of the following function:

$$F(s) = \frac{100s(s^2 + 30s + 400)}{(s+25)(s^3 + 6s^2 + 16s + 16)}$$

9-30 Use MATLAB to find the inverse transform and plot the poles and zeros of the following function:

$$F(s) = \frac{50(s^3 + 2s^2 + s + 2)}{s(s^3 + 4s^2 + 4s + 16)}$$

OBJECTIVE 9-3 CIRCUIT RESPONSE USING LAPLACE TRANSFORMS (SECT. 9-6)

Given a first- or second-order circuit:

- (a) Determine the circuit differential equation and the initial conditions (if not given).
 (b) Transform the differential equation into the s domain and solve for the response transform.
 (c) Use the inverse transformation to find the response waveform.

See Examples 9-15, 9-16, 9-17, 9-18 and Exercises 9-22, 9-23, 9-24, 9-25

9-31 Use the Laplace transformation to find the $v(t)$ that satisfies the following first-order differential equations:

- (a) $500 \frac{dv(t)}{dt} + 2500v(t) = 0, \quad v(0-) = 100 \text{ V}$
 (b) $\frac{dv(t)}{dt} + 200v(t) = 400u(t) \text{ V}, \quad v(0-) = -100 \text{ V}$

9-32 Use the Laplace transformation to find the $i(t)$ that satisfies the following first-order differential equation:

$$\frac{di(t)}{dt} + 500i(t) = [0.100e^{-200t}]u(t) \text{ A}, \quad i(0-) = 0 \text{ A}$$

9-33 The switch in Figure P9-33 has been open for a long time and is closed at $t = 0$. The circuit parameters are $R = 400 \Omega$, $L = 330 \text{ mH}$, and $V_A = 15 \text{ V}$.

- (a) Find the differential equation for the inductor current $i_L(t)$ and the initial condition $i_L(0)$.
 (b) Solve for $i_L(t)$ using the Laplace transformation.

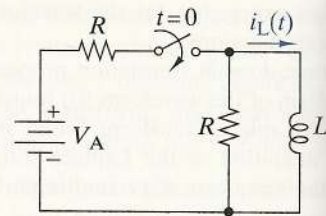


FIGURE P9-33

9-34 The switch in Figure P9-33 has been open for a long time and is closed at $t = 0$. The circuit parameters are $R = 200 \Omega$, $L = 200 \text{ mH}$, and $V_A = 50e^{-1000t} \text{ V}$.

- (a) Find the differential equation for the inductor current $i_L(t)$ and initial condition $i_L(0)$.
 (b) Solve for $i_L(t)$ using the Laplace transformation.

9-35 The switch in Figure P9-35 has been open for a long time and is closed at $t = 0$. The circuit parameters are $R = 8 \text{ k}\Omega$, $C = 125 \text{ nF}$, and $V_S(t) = 25u(t) \text{ V}$.

- (a) Find the differential equation for the capacitor voltage $v_C(t)$ and initial condition $v_C(0)$.
 (b) Find $v_C(t)$ using the Laplace transformation.

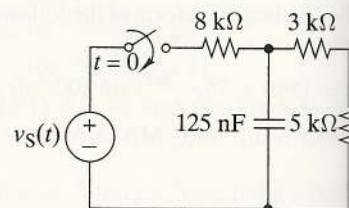


FIGURE P9-35

9-36 Repeat Problem 9-35 for the input voltage $v_S(t) = 100[\cos 1000t]u(t) \text{ V}$.

9-37 Use the Laplace transformation to find the $v(t)$ that satisfies the following second-order differential equation:

$$\frac{d^2v(t)}{dt^2} + 25 \frac{dv(t)}{dt} + 100v(t) = 0, \quad v(0-) = 10 \text{ V and } \frac{dv(0-)}{dt} = 0$$

9-38 Use the Laplace transformation to find the $v(t)$ that satisfies the following second-order differential equation:

$$\frac{d^2v(t)}{dt^2} + 100 \frac{dv(t)}{dt} + 500v(t) = 0, \quad v(0-) = 0 \text{ and } \frac{dv(0-)}{dt} = 100 \text{ V/s}$$

9-39 The switch in Figure P9-39 has been open for a long time and is closed at $t = 0$. The circuit parameters are $R = 500 \Omega$, $L = 2.5 \text{ H}$, $C = 2.5 \mu\text{F}$, and $V_A = 20 \text{ V}$.

- (a) Find the circuit differential equation in the s domain and the initial conditions $i_L(0)$ and $v_C(0)$.
 (b) Use Laplace transforms to solve for $i_L(t)$.